

Labor and Life: Reassessing the Employment–Fertility Relationship Worldwide

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ABSTRACT

The decline in fertility rates around the world is alarming as the situation is getting worse over time (Gietel-Basten et al.). Among the OECD nations, South Korea had the lowest fertility rate of 0.81 births per woman in 2021 (OECD). To figure out what factors are associated with the fertility rate, I conducted research to figure out the question of whether there is a correlation between the employment-to-population ratio and fertility rate. From my research, I found a non-linear relationship between the employment-to-population ratio and the fertility rate. The employment-to-population ratio, GDP per capita, urban population, and control of corruption had a negative coefficient, which means that one unit of each variable is associated with a decrease in a certain number of births per woman. This study's objective is to find out which variables are affecting the actual fertility rate, and it found out gross savings and real interest rate are not associated with the fertility rate.

I. INTRODUCTION

The main purpose of this research paper is to find the relationship between the employment-to-population ratio and women's fertility rate. From the research paper that investigates the relationship between the fertility rate and mother's employment from the period after World War II, having children had a negative effect on female employment (Julio). Conversely, the developing countries showed an opposite causal relationship.

This paper will introduce the confounding variables along with the main variables: gross savings(adjusted savings), GDP per capita, urban population, and several other variables that may be associated with the fertility rate. According to Ahn and Mira's paper, the research team found a reversal relationship between the female participation rate and the total fertility rate. The correlation between the two got higher after 1985 (Ahn and Mira). In my paper, I would like to bring up the employment-to-population ratio instead of the particular female employment rate because baby care is usually driven by both men and women, so I thought it would target the household's perspective, not just for women.

II. LITERATURE REVIEW

One research paper introduces 23 OECD countries, using panel data that includes the period between 1960 and 1997. From this paper, the final model found a "strong negative and highly significant coefficient for female unemployment" (Adsera). This means that a one percent increase in the unemployment rate is associated with a decline in the fertility rate.

The financial or nationwide crisis also has an impact on the fertility rate. The report from the International Monetary Fund(IMF) released the paper about economic recovery after the 2008 US Financial Crisis. IMF stated that the economic crisis will lead to an increase in the unemployment rate, and mothers to delay childbearing which is possible to reduce the short-term

fertility rate (Chen et al.). The crisis' aftermath continued to impact the nation's economic performance, especially in migration and fertility. Then, the reduced fertility rate influenced GDP losses, GDP per capita losses, and employment losses. Among the three, the employment losses were most significant in both the short-term and medium-term.

The other source said that employment and income for each person have a strong impact on a mother's time (Ahn and Mira). The rise in wages increased the fertility rate, and the countries with high female participation rates didn't bear babies but started to recover in the early 1980s.

III. DATA DESCRIPTION

This research utilized panel data at the country level with 217 nations from 2005 to 2022 by the World Bank Data Bank. The total observations (including countries that have missing values in some variables) are 3,906.

Table 1: Variable Descriptions

<i>fertrate</i>	Female Fertility Rate (birth per woman)
<i>employtopop</i>	Employment-to-Population Ratio (age 15+, in %)
<i>grosssav</i>	Adjusted Savings: Gross Savings (in % of GNI)
<i>GDPpercap_in_thousand</i>	GDP per Capita in PPP (constant 2017 international \$, in \$1,000)
<i>realintrate</i>	Real Interest Rate (in %)
<i>urban_in_tenthousand</i>	Urban Population (in ten thousand)
<i>femaleedu</i>	Female Educational Attainment (Bachelor or equivalent and above, age 25+, in %)
<i>ctrlcorrupt</i>	Rate of Control of Corruption (in %)

From Table 2 on the next page, the dependent variable, *fertrate*, represents the country's female fertility rate. I first tried to gather the data from the OECD, but it was not enough that there were only 38 OECD countries. The data reported was better in the OECD, but decided to

Table 2: Summary Statistics

VARIABLES	(1) N	(2) mean	(3) sd	(4) min	(5) max
fertrate	3,580	2.782	1.431	0.772	7.615
employtopop	3,365	56.15	11.42	22.63	88.52
grosssav	2,703	23.13	12.10	-29.26	75.76
GDPpercap_in_thousand	3,456	20.74	22.05	0.708	157.6
realintrate	2,311	5.464	9.575	-81.13	61.88
urban_in_tenthousand	3,870	1,791	6,346	0.493	89,758
femaleedu	574	17.05	11.11	0	60.98
ctrlcorrupt	3,656	49.15	29.06	0	100

get data from the World Bank because it comprises every country which also includes nations that are not in the OECD.

The *employtopop* represents the employment-to-population ratio, which includes all working employed citizens over the total working-age population. The *grosssav* represents the subtraction from consumption to disposable income. The *GDPpercap_in_thousand* is the variable derived from GDP per capita, PPP that was standardized by constant 2017 international dollar. Due to getting numbers with a scientific notation that is hard to interpret, I converted the unit to \$1,000. *Realintrate* represents the nation's real interest rate, which I thought would be helpful to indicate the market price of the country. *Urban_in_tenthousand* was released with a pure number of the urban population. However, I converted the unit to ten thousand due to hard interpretation. *Femaleedu* was gathered to know what portion of females have higher education which may affect employment. The last variable *ctrlcorrupt* represents how the nationals can handle the corruption in the country. The higher the index, the higher the countries can cope with the corruption. This was gathered to indicate how the countries are stable internally which I thought can influence the fertility rate.

IV. EMPIRICAL MODEL

$$\begin{aligned} \text{fertrate}_{it} = & \beta_0 + \beta_1 \text{employmenttopop}_{it} + \beta_2 \text{grosssav}_{it} \\ & + \beta_3 \text{GDPpercap}_{it} + \beta_4 \text{employmenttopop}_{it} * \text{GDPpercap}_{it} + \beta_5 \text{realintrate}_{it} \\ & + \beta_6 \text{urbanpop}_{it} + \beta_7 \text{femaleedu}_{it} + \beta_8 \text{ctrolcorrupt}_{it} + Z_i + T_t + u_{it} \end{aligned}$$

As this paper uses panel data, fixed effects are required. So i will represent country-fixed effects, and t will represent the time-fixed effects. The hypothesized model starts with the linear regression of fertility rate on employment to population ratio, gross savings, GDP per capita, real interest rate, urban population, female education attainment rate, and the rate of control of corruption. From the literature, I hypothesize that the employment-to-population ratio will have a negative impact on the fertility rate and that higher employment will lead to a decrease in the fertility rate. In addition, I guessed the GDP per capita influences the employment-to-population ratio in one country, so I tried to apply the interaction term.

Figure 1 is a scatterplot of the relationship between employment to population ratio to fertility rate for 217 countries. It limits the year to 2009, and the reason the graph is limited to 2009 is because the recent year 2022 doesn't have sufficient value. 2009 had a sufficient amount of data, and I think it was interesting as 2009 was after the US financial crisis in 2008. From this scatterplot, Niger, Somalia, Chad, and Djibouti are noticeable outliers compared to the center where most of the countries are located.

Figure 2 is a scatterplot of the relationship between GDP per capita(in \$1,00) and fertility rate. The reason why this paper includes the graph of scatterplot between GDP per capita and fertility rate is because I wanted to know how output per person in one country is associated with the country's fertility rate. However, the figure is limited to a total of 20 countries, split by the top 10 countries' GDP per capita and the lowest 10 countries' GDP per capita. However, due to some countries (San Marino, and the Cayman Islands) missing reports on their fertility rate, I

pulled in the next two countries (Norway, and the United Arab Emirates) that have high GDP per capita.

Figure 1:

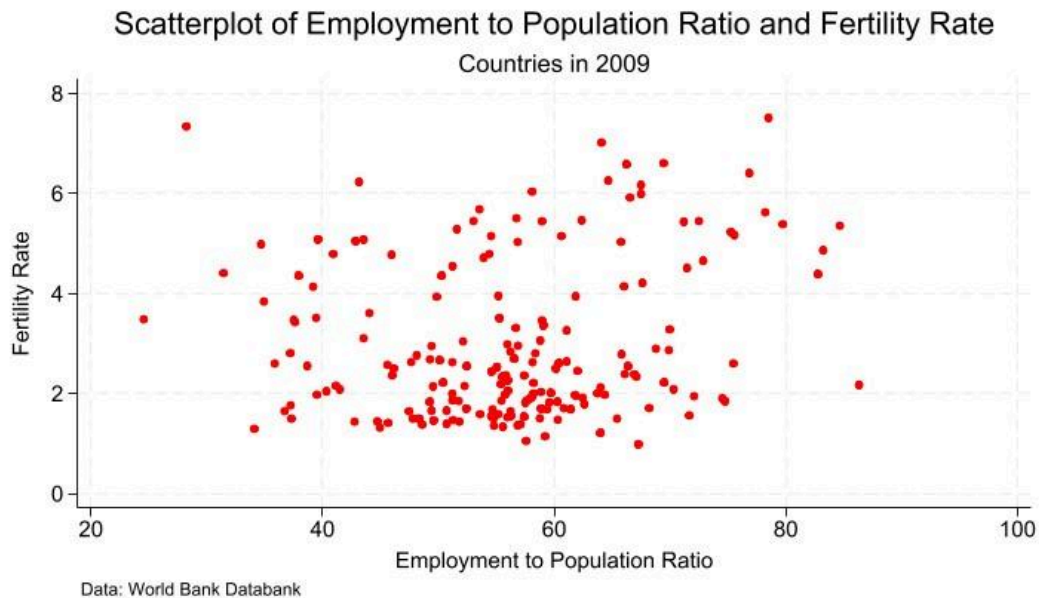
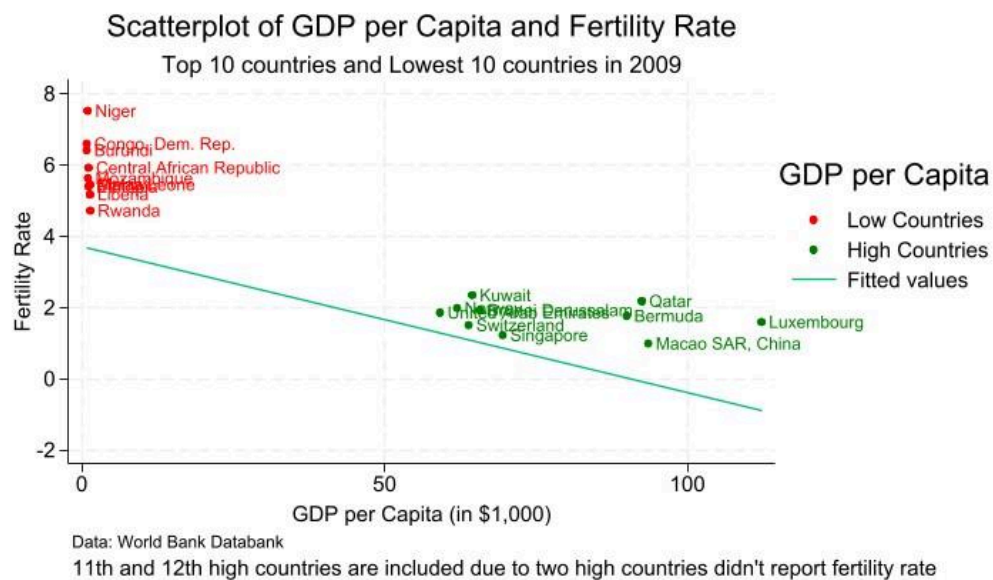


Figure 2:



V. RESULTS

Table 3: Regression Results

VARIABLES	Regression Results					
	(1) fertrate	(2) fertrate	(3) fertrate	(4) fertrate	(5) fertrate	(6) fertrate
employtopop	0.0233* (0.0120)	0.0418*** (0.0114)	0.0337*** (0.0118)	-0.120** (0.0575)	0.0359*** (0.00879)	0.0320*** (0.00655)
grosssav		-0.0159* (0.00944)				
GDPpercap_in_thousand		-0.0282*** (0.00663)	-0.0711** (0.0288)	-0.0217 (0.0253)	0.0368** (0.0151)	0.0570*** (0.0156)
realintrate		-0.00555 (0.0119)				
urban_in_tenthousand		-2.42e-05*** (8.68e-06)		-3.27e-05*** (8.09e-06)	-3.71e-05 (2.72e-05)	
ctrlcorrupt		-0.0152*** (0.00502)		-0.0170*** (0.00443)	0.000566 (0.00259)	
employment_gdp			0.000427 (0.000403)	-6.67e-05 (0.000336)	-0.000614*** (0.000235)	-0.000722*** (0.000209)
employment_sqr				0.00139*** (0.000498)		
2009.Time						-0.0861*** (0.0165)
Constant	1.693** (0.674)	2.172*** (0.657)	1.942*** (0.711)	6.520*** (1.604)	0.848 (0.514)	0.981** (0.397)
Observations	187	108	173	173	2,959	2,959
Adjusted R-squared	0.024	0.421	0.381	0.470	0.087	0.458
Country FE	NO	NO	NO	NO	YES	YES
Year FE	NO	NO	NO	NO	NO	YES
Number of country_num					175	175

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 3 starts with the simple linear regression that controls only for the main independent variable *employtopop*. Model 1 can be interpreted as a one percentage point increase in the employment-to-population ratio is associated with a 0.0233 increase in births per woman. I ran the regression setting 2009, and throughout this regression, I will set the year as 2009, except for using fixed effects, which are Model 5 and Model 6.

Model 2 is the model that included all the variables that I had in the dataset, which includes *femaleedu* which are not listed in this table. This model can be interpreted as one percentage point increase in the employment-to-population ratio is associated with a 0.0418 increase in birth per woman when we hold other variables constant.

Before getting to the final Model 2, there was collinearity in *GDPpercap_in_thousand* and *femaleedu*. So I ran two regressions: one that excludes *GDPpercap_in_thousand* but includes all the other variables, and the other excludes *femaleedu* but includes all the other variables. As a result, the regression that excludes *femaleedu* had only six observations available when I restrict the year to 2009. So I decided not to use *femaleedu* from now on. As a result, *realinrate* became the only variable that is insignificant. The variable *grosssav* is significant at the 10% level. *employtopop*, *GDPpercap_in_thousand*, *urban_in_tenthousand*, and *ctrlcorrupt* are significant at the 1% level. As *realinrate* is insignificant, I decided to no longer insert this variable in future models.

Model 3 is the model that includes interaction terms. Before controlling the interaction term with *employtopop* and *GDPpercap_in_thousand*, I tried running the interaction model between *employtopop* and *urban_in_tenthousand*. When I ran the model, the interaction term didn't turn out significant, and *employtopop*'s coefficient didn't change more than half a standard error. So I tried the interaction terms that I thought in the hypothesized model. As a result, I found out that though the interaction term didn't have a statistically significant feature, the coefficient changed more than half the standard error. We can say that the effect of employment to population ratio on the fertility rate depends on *GDPpercap_in_thousand*. The coefficient of *employtopop* now represents the effect of a one percentage point increase in employment to

population ratio is associated with a 0.0337 increase in birth per woman if GDP per capita is zero.

Model 4 introduces the non-linear relationship. This model shows the highest Adjusted R^2 , which I can interpret this model as the variation of fertility rate by employment-to-population ratio can be explained with 47%. I conducted a 10-step process and compared the regression of *fertrate* on *employtopop* and the regression of $\ln(\text{fertrate})$ on *employtopop*. As a result, the former R^2 can be interpreted as 1.84% of the variation in fertility rate can be explained by the employment-to-population ratio, while the latter R^2 can be interpreted as 2.19% of the variation in fertility rate is explained by the employment-to-population ratio. As the latter relationship has higher R^2 , I chose this as the model. Next, I compared the regression of $\ln(\text{fertrate})$ on *employtopop* and the regression of *fertrate* on *employtopop* and *employtopop*². To compare these two, I needed an Adjusted R^2 . As a result, I got 0.0166 and 0.1242, respectively. So I dropped $\ln(\text{fertrate})$ and decided to use *fertrate* with a quadratic model. As we are allowing a quadratic model, the changes in fertility rate will differ on the right and left sides of the turning point, which is 43.17% on *employtopop*. It shows the negative and diminishing effects of *employtopop* on *fertrate*. Though the *GDPpercap_in_thousand* and *employment_gdp* showed insignificance, I decided to leave it in the model because the coefficient changed more than half a standard error.

Model 5 introduces the country fixed effects model. It will control for all time-invariant factors that vary across countries. I first ran the model including *employment_sqr*, but I decided to get rid of the squared term because of insignificance and reduced Adjusted R^2 . When I decided to keep the interaction term, I found the Adjusted R^2 increased, and the coefficients

remained significant. However, as soon as I pulled in the *urban_in_tenthousand* and *ctrlcorrupt*, the p-value showed higher than 0.1, which was insignificant. After running a joint F-test, it showed a p-value of 0.3873, which was insignificant. When we are holding *GDPpercap_in_thousand*, *urban_in_tenthousand*, *ctrlcorrupt* constant, a one percent point increase in the employment-to-population ratio is associated with a 0.0359 increase in birth per woman. There is a turning point -58.46 at *GDPpercap_in_thousand*. However, as *GDPpercap_in_thousand*'s minimum value is 0.708, we can ignore the turning point.

Model 6 introduces both time and country-fixed effects. This model will show that the variables vary across countries within a year, and vary across year within a country. This model shows variation of fertility rate by employment-to-population ratio can be explained with 45.8%. Due to the insignificant result from the previous model's joint F-test, I decided to run only with the main independent variable, *GDPpercap_in_thousand*, and the interaction term. As a result, the main independent variable turned out to be significant at a 1% level, and it can be interpreted as a one percentage point in the employment-to-population ratio associated with a 0.0320 increase in birth per woman. The final model reveals that there's no change in the slope on *employtopop*. As the coefficient of *employtopop* is positive and *employment_gdp* is negative with both being significant, we can say the relationship between *employtopop* and *fetrte* starts out positive, but it can turn negative past the turning point, which is -44.32. Again, we can ignore the turning point here as the *GDPpercap_in_thousand*. As all the data will be on the right side from the turning point, we can say *employtopop* and *fetrte* have a negative relationship.

VI. CONCLUSION

From the best model - Model 4 - we can compare from the hypothesized model that the best model shows the gross savings and real interest rate is not associated with fertility rate. The

employment-to-population ratio is significant at a 5% level, and the squared employment-to-population ratio is significant at a 1% level. The employment-to-population ratio of 43.17% indicates the turning point. The left part of the turning point will show a declining fertility rate, while the fertility rate will start to go higher from the right part of the turning point. This can be interpreted as the fertility rate will increase as the employment-to-population ratio exceeds 43.17%. The interaction term turned out insignificant, but it reduced the employment-to-population ratio's bias. It was interesting that the country's control of corruption influenced the fertility rate. With the significance of a 1% level on the coefficient of control of corruption, the government's role is important, and this can include the authority's active role in governance and policy-making.

VII. LIMITATIONS AND FUTURE RESEARCH

The limitations of this research are that we didn't control for social factors, such as competitiveness, culture, and some miscellaneous elements that make it hard to get data as numbers and have no clear standards. In addition, the female education attainment rate was not enough to regress in this research. Future research should include more data on the female education attainment rate.

Gietel-Basten's paper introduces "target-driven pronatalism," which is a policy to stimulate marriage and larger families. This paper says the fertility rate increment should be followed by policies such as broadening access to health and social policies. If there is a chance to analyze the policy of fertility with economic analysis in certain nations, it would be helpful to specify what each country implements the fertility policy and combine the social factors, which can be one of the standards that policymakers consider before implementing the policies.

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